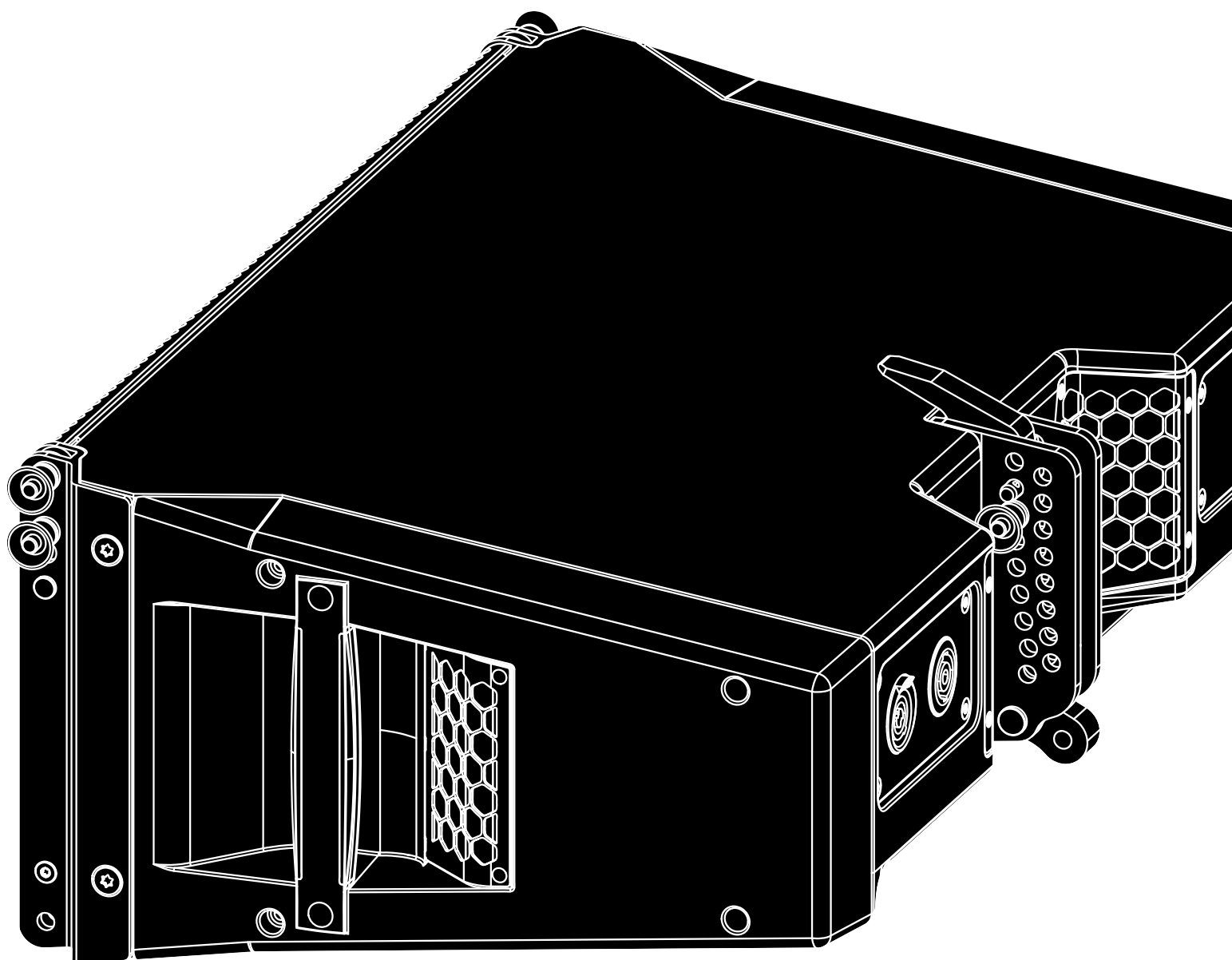


CL

CCL8/CCL12
Manual 1.5 en



Notes on document version

All previous versions of this document are hereby no longer valid.

Version 1.5:

Revised Max. SPL data as well as equivalent total LF driver cone size added.

Refer to:

⇒ Chapter 2.5 "Technical specifications" on page 9.

General information

CCL8/CCL12 Manual

Version: 1.5 en, 02/2026, D2780.EN .01

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d&b audiotechnik GmbH & Co. KG
Eugen-Adolff-Str. 134, D-71522 Backnang, Germany
T +49-7191-9669-0, F +49-7191-95 00 00

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Potential risk of personal injury

Never stand in the immediate vicinity of loudspeakers driven at a high level. Professional loudspeaker systems are capable of causing a sound pressure level detrimental to human health. Seemingly non-critical sound levels (from approx. 95 dB SPL) can cause hearing damage if people are exposed to it over a long period.

In order to prevent accidents when deploying loudspeakers on the ground or when flown, please take note of the following:

- When setting up the loudspeakers or loudspeaker stands, make sure they are standing on a firm surface. If you place several systems on top of one another, use straps to secure them against movement.
- Only use accessories which have been tested and approved by d&b for assembly and deployment. Pay attention to the correct application and maximum load capacity of the accessories as detailed in our specific "Mounting instructions" or in our "Rigging manuals".
- Ensure that all additional hardware, fixings and fasteners used for deployment are of an appropriate size and load safety factor. Pay attention to the manufacturers' instructions and to the relevant safety guidelines.
- Regularly check the loudspeaker housings and accessories for visible signs of wear and tear, and replace them when necessary.
- Regularly check all load bearing mounting devices.

Potential risk of material damage

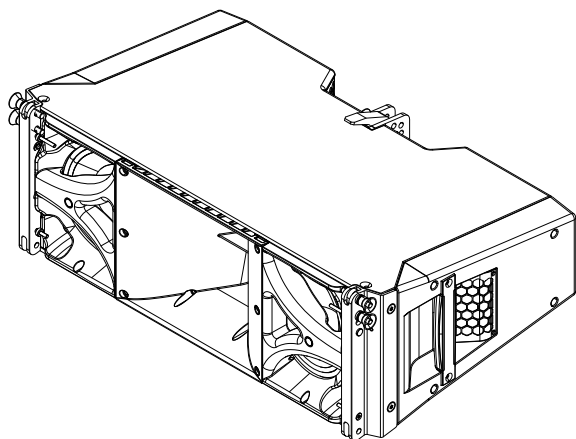
Loudspeakers produce a static magnetic field even if they are not connected or are not in use. Therefore make sure when erecting and transporting loudspeakers that they are nowhere near equipment and objects which may be impaired or damaged by an external magnetic field. Generally speaking, a distance of 0.5 m (1.5 ft) from magnetic data carriers (floppy disks, audio and video tapes, bank cards, etc.) is sufficient.

NOTICE!

Only operate d&b loudspeakers with correctly configured d&b amplifiers, otherwise there is a risk of damaging the loudspeaker components and the directional characteristics of the system cannot be achieved.

Recommended amplifiers

D25/D40/D90/D80



2.1 Intended use

NOTICE!

Possible risk of water ingress!

The passive LF cardioid design of the d&b CL-Series systems requires dedicated rear ports. For acoustic reasons, these ports cannot be fully water sealed.

When the system is deployed or installed in inclement weather with heavy rain, water ingress may occur.

While permanent damage is unlikely under normal conditions, water ingress can require cabinet disassembly for drainage to prevent damage to internal components.

Users must exercise caution and take special precautions during outdoor application in bad weather conditions with heavy rainfall. In such situations, the systems must be deployed underneath a roof or a roof overhang, whenever possible.

If direct exposure to severe precipitation cannot be avoided, the rear of the array must be protected using suitable means (custom rain covers, tarpaulins, or plastic wraps).

Product description

The CCL8 is a compact cardioid line array loudspeaker for small to medium-scale sound reinforcement in mobile applications. When the CCL Flying frame is used, up to 24 cabinets can be flown in vertical columns producing an 80° constant directivity dispersion pattern in the horizontal plane.

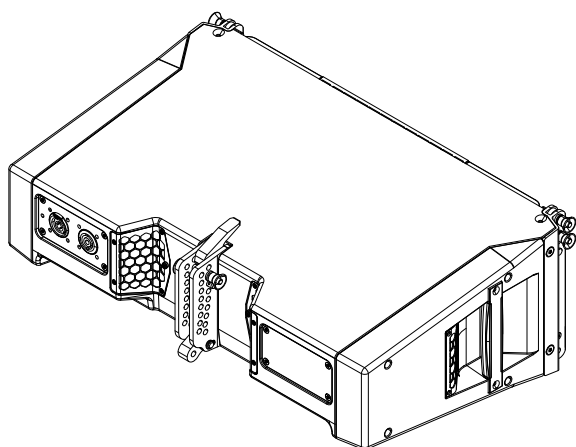
The CCL12 line array module is acoustically and mechanically compatible with the CCL8 and provides a 120° horizontal dispersion.

The cabinets are 2-way passive designs housing 2 x 7" neodymium forward LF drivers, 2 x 5" neodymium side firing LF drivers and two 1.75" HF compression drivers mounted to a dedicated wave shaping device. The cylindrical wave segments of each cabinet couple without gaps and sum up coherently. Splay angles between adjacent cabinets can be set in the range from 0° to 14° in 1° increments.

All components are arranged symmetrically around the center axis of the cabinet to produce a perfect symmetrical dispersion pattern. This setup allows for a crossover design with a well defined overlap of adjacent frequency bands resulting in a very consistent and accurate horizontal dispersion.

Due to the arrangement of the forward and sideward LF drivers in combination with their processing functions, the directivity is maintained across the entire operating bandwidth.

The frequency response extends from 60 Hz to above 18 kHz.

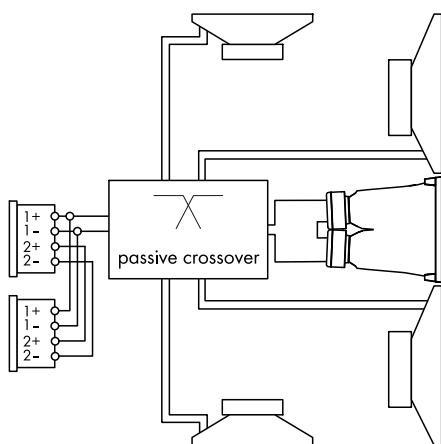


The cabinet enclosure is injection molded (ABS Polycarbonate) and has an impact and weather protected 2K finish. The front baffle as well as the side and rear ports of the cabinet incorporate a rigid metal grill backed by an acoustically transparent and water repellent fabric. Each side panel incorporates a handle while two additional support handles are provided at the rear.

CL-Series rigging components and arrays

The cabinets are mechanically connected using the rigging strands on both sides of the cabinet front and a central strand at the rear of the cabinet. All necessary rigging components are mounted on the cabinet and fold out or slide out when needed.

A detailed description of the rigging components is given in the respective CCL Rigging manual.



NLT4 F/M Connector wiring

2.2 Connections

The cabinets are fitted with NLT4 F/M connectors. All four pins of both connectors are wired in parallel. The cabinets use the pin assignments 1+/1-. Pins 2+/2- are designated to subwoofers. Using the male connector as the input, the female connector allows for direct connection to a second cabinet.

d&b LoadMatch

With the d&b four channel amplifier platform, the LoadMatch function enables the amplifier to electrically compensate for the properties of the loudspeaker cable used without the need for an additional sense wire. For applicable loudspeakers, LoadMatch is therefore independent of the connector type used.

2.3 Operation

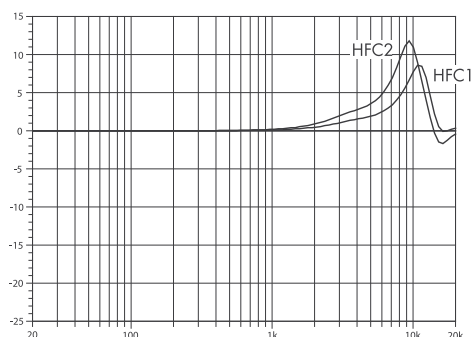
Amplifier output mode(s): Dual Channel or Mix TOP/SUB		
Application	Setup	Cabinets per channel
CCL8	CCL8 Line	2
	CCL8 Arc	2
	CCL8 AP	1
CCL12	CCL12 Line	2
	CCL12 Arc	2
	CCL12 AP	1

Line and Arc setups

The selection of "Line" or "Arc" depends on the curvature of the array. Both setups may be used within one array.

The "Line" setup is used for long throw array sections with three or more consecutive splay settings of 0°, 1° or 2°. Compared to the "Arc" setup, the mid/high range is reduced to compensate for the extended nearfield.

The Arc setup is intended for line array loudspeakers when used in curved array sections.



Frequency response correction of the HFC function *

* schematic diagram

AP setup

In connection with d&b ArrayProcessing (AP), the AP setup contains the AP data that is generated by the d&b ArrayCalc simulation software and transferred to the applicable amplifiers via the d&b Remote network (OCA/AES70) using R1.

As soon as the data has been sent to the amplifiers, the AP setup is automatically activated.

2.3.1 Controller settings

For acoustic adjustment the functions CUT, HFC and Coupling can be selected.

CUT mode

Set to CUT, the low frequency level of the cabinets is reduced. The CCL8/CCL12 cabinets are now configured for use with applicable d&b subwoofers such as the CCL-SUB.

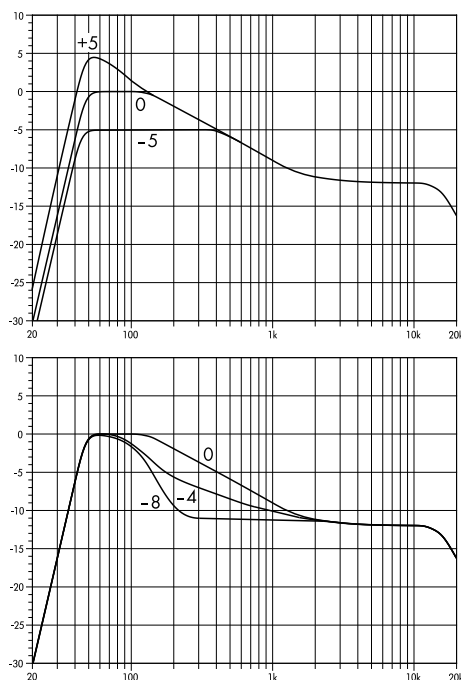
HFC function (Line/Arc setups only)

Selecting the HFC (High Frequency Compensation) function compensates for the loss of high frequency energy due to air absorption when loudspeakers are used to cover far field listening positions.

The HFC function has two settings (HFC1 and HFC2) for different distances the cabinets need to cover. The settings should be used selectively; HFC1 compensates for 40 m (131 ft) and HFC2 for 80 m (262.5 ft) of additional distance from a reference position.

The compensation is adjusted for a typical relative humidity of 50 % at 22 °C. With lower humidity the absorption by air increases, therefore the distances where the respective HFC setting provides a correct equalization are shorter than indicated above.

Using the HFC function provides the correct sound balance between close and remote audience areas, whilst all amplifiers driving the array can be fed with the same signal.



Frequency response correction of the Coupling function for low and low-mid frequency level*

* schematic diagram

Coupling function

The Coupling function compensates for coupling effects between the cabinets by reducing the low and low-mid frequency level.

The function provides a two-stage filter (Low/Mid), which allows the independent shaping of the low and low-mid responses.

The characteristics of the Coupling function are shown in the diagram opposite. The standard setting (0) maintains the default array response. Coupling values can be set in the range from +5 to -5 (Low) and from 0 to -8 (Mid) in increments of 0.5.

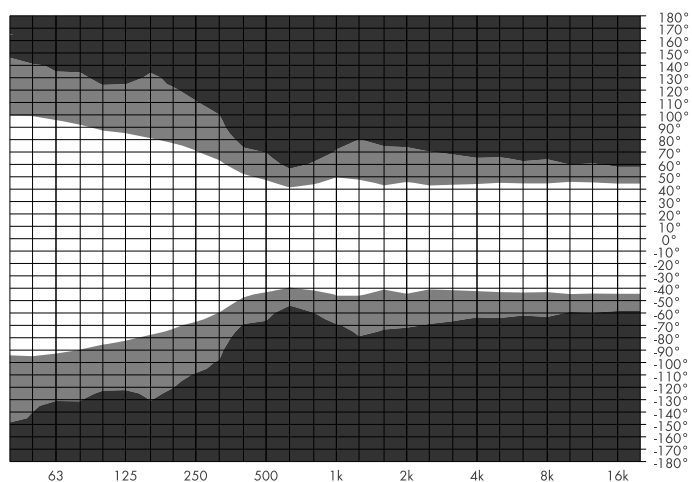
Note: Please note that all cabinets within the array should be operated with the same Coupling setting.

When processed conventionally (Line/Arc), the larger the array the more attenuation by Coupling will be required to achieve a neutral response.

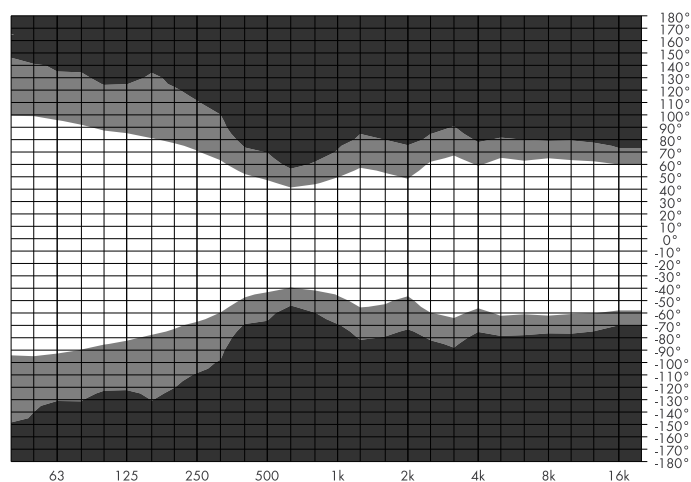
When operated with ArrayProcessing (AP), an array will automatically be provided with the system target response, as shown in the graphic opposite. All coupling effects caused by the array length and shape are considered in the AP data. The Coupling function may still be used for additional corrections, for example of room properties or coupling effects between main hangs and outfills.

2.4 Dispersion characteristics

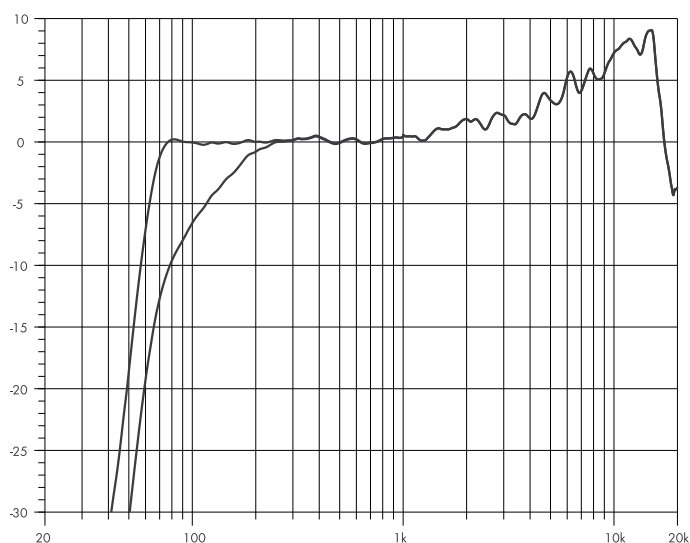
The graphs below show the horizontal dispersion angle over frequency plotted using lines of equal sound pressure (isobars) at -6 dB and -12 dB. The nominal horizontal dispersion of 80° / 120° is maintained over the entire operating bandwidth.



CCL8 isobar diagram horizontal

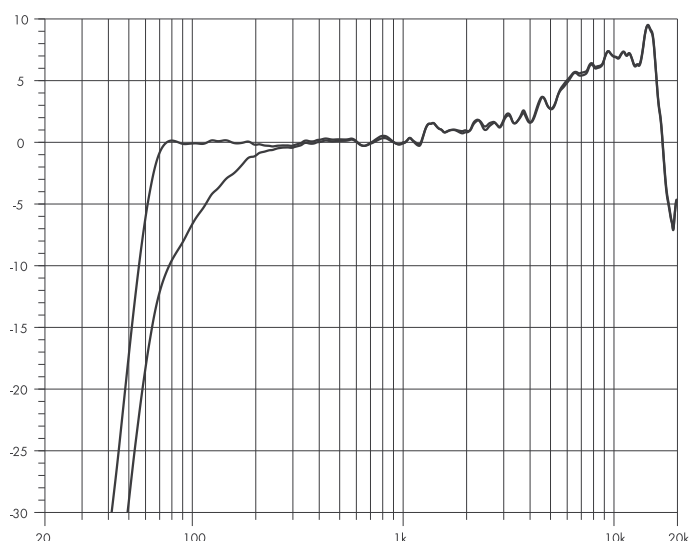


CCL12 isobar diagram horizontal



CCL8 Frequency response*, Standard and CUT modes

*single cabinet within array



CCL12 Frequency response*, Standard and CUT modes

*single cabinet within array

2.5 Technical specifications

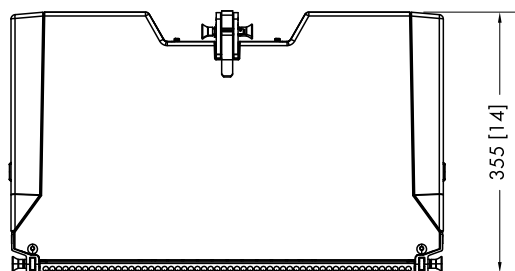
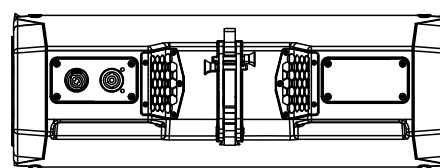
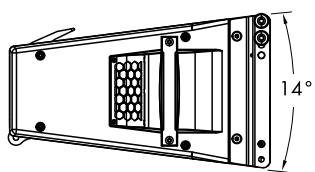
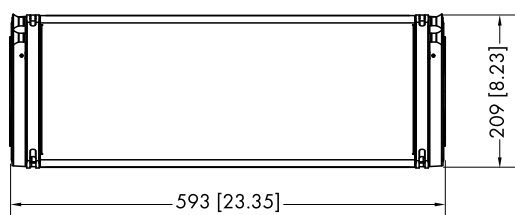
System data

Frequency response (-5 dB standard)	60 Hz - 18 kHz
Frequency response (-5 dB CUT mode)	90 Hz - 18 kHz
Frequency response (-10 dB standard, IEC60268)	55 Hz - 20 kHz
Frequency response (-10 dB CUT mode, IEC60268)	80 Hz - 20 kHz
Max. SPL (Peak, 1 m, free field, broadband signal IEC60268)	
CCL8 with D25/D20/25D/30D	135 dB
CCL8 with D90/D80/D40/40D	137 dB
4x CCL8 (far field, scaled to 1 m)	
with D90/D80/D40/40D	149 dB
CCL12 with D25/D20/25D/30D	134 dB
CCL12 with D90/D80/D40/40D	136 dB
4x CCL12 (far field, scaled to 1 m)	
with D90/D80/D40/40D	148 dB

Loudspeaker data

Nominal impedance	10 ohms
Power handling capacity (RMS/peak 10 ms)	
	400/1200 W
Nominal horizontal dispersion angle	
CCL8	80°
CCL12	120°
Splay angle setting	0 ... 14° (1° increment)
Components	2x 7" front LF driver
	2x 5" side LF driver
	(2x 8.7" total equivalent* LF driver size)
	2x 10.2 x 63 mm exit HF compression drivers with 1.75" coil
	Passive crossover network
Connections	NLT4 F/M
Pin assignment	1+/1-
Weight	17.6 kg (38.8 lb)

* Calculated equivalent diameter of coherent addition of front and side radiating LF drivers.



CCL8/CCL12 cabinet dimensions in mm [inch]



3.1 Conformity of loudspeakers

This declaration applies to:

- **d&b Z0880 CCL8 loudspeaker**
- **d&b Z0882 CCL12 loudspeaker**

by d&b audiotechnik GmbH & Co. KG.

All product variants are included, provided they correspond to the original technical version and have not been subject to any later design or electromechanical modifications.

We herewith declare that said products are in conformity with the provisions of the respective directives including all applicable amendments.

Detailed and applicable declarations are available on request and can be ordered from d&b or downloaded from the d&b website at www.dbaudio.com.



3.2 WEEE Declaration (Disposal)

Electrical and electronic equipment must be disposed of separately from normal waste at the end of its operational lifetime.

Please dispose of this product according to the respective national regulations or contractual agreements. If there are any further questions concerning the disposal of this product, please contact d&b audiotechnik.

WEEE-Reg.-Nr. DE: 13421928

